
Lab Project: Deploy Docker on Amazon Linux 2 EC2 and Run a Web App in a Docker Container

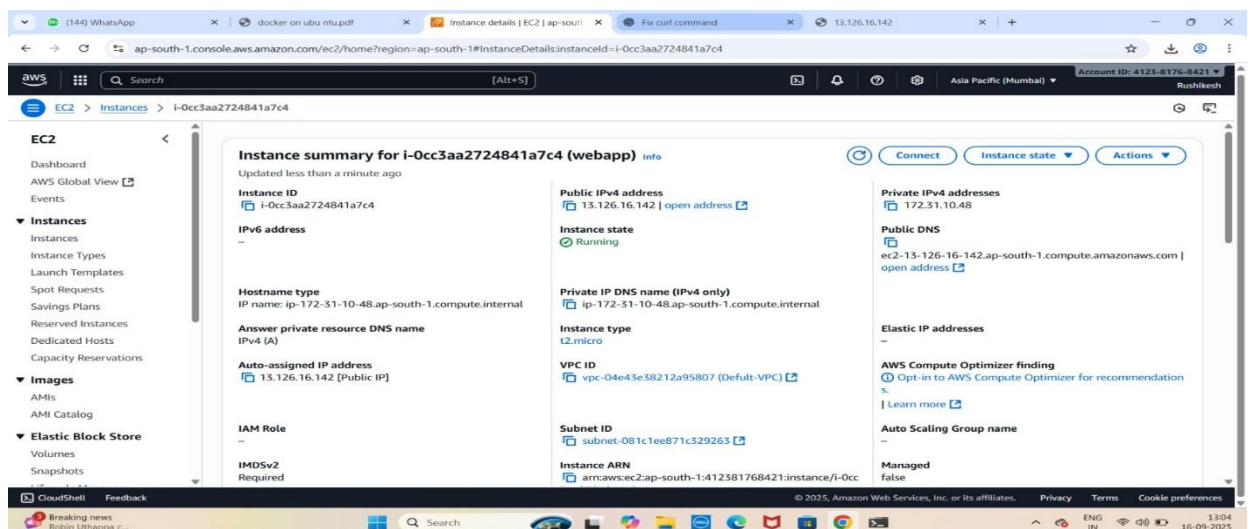
Objective

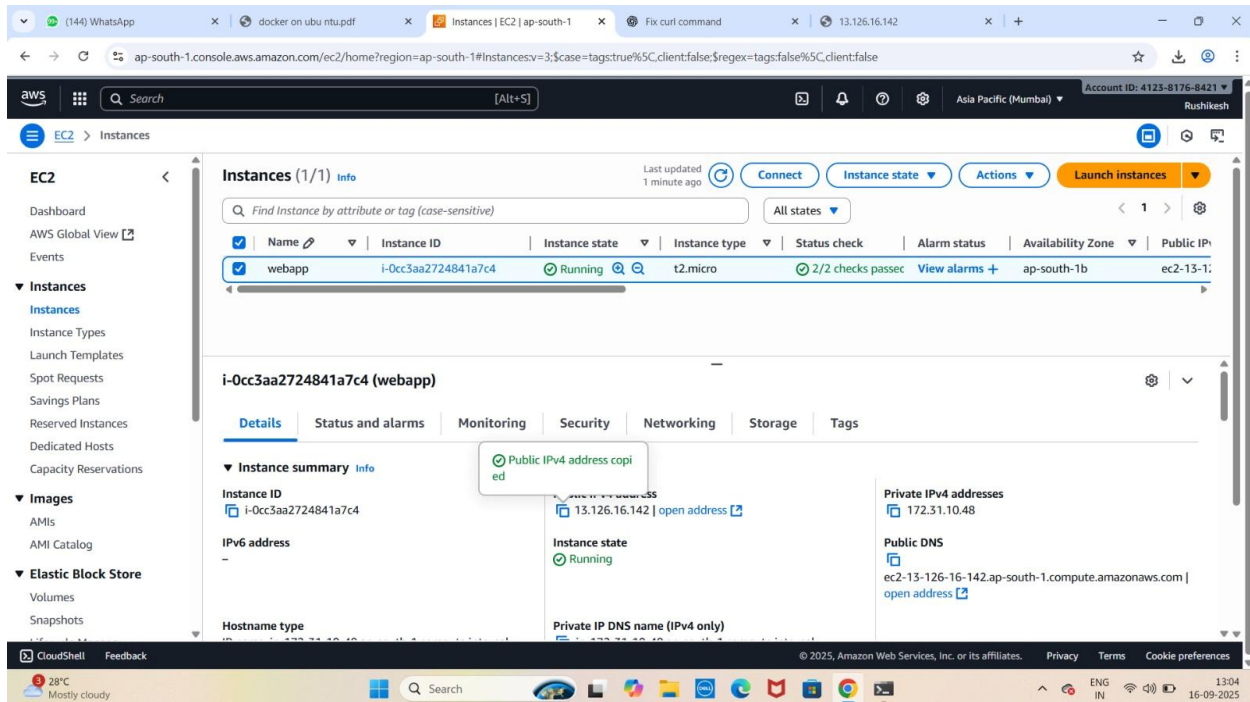
By the end of this lab, you will:

- Launch an **Amazon Linux 2 EC2 instance**.
 - Install **Docker** on it.
 - Run an **Apache (httpd) container**.
 - Serve a **custom webpage** from within the Docker container.
-

Step 1: Launch EC2 Instance

1. Go to **AWS Management Console** → **EC2** → **Launch Instance**.
2. Enter name: **AmazonLinux2-Docker-Lab**.
3. Choose **AMI**: Amazon Linux 2 (64-bit x86).
4. Select **Instance Type**: t2.micro (Free Tier eligible).
5. **Key Pair**: Choose or create a new .pem key.
6. **Security Group**: Allow **SSH (22)** and **HTTP (80)**.
7. Launch the instance and copy the **Public IPv4 Address**.

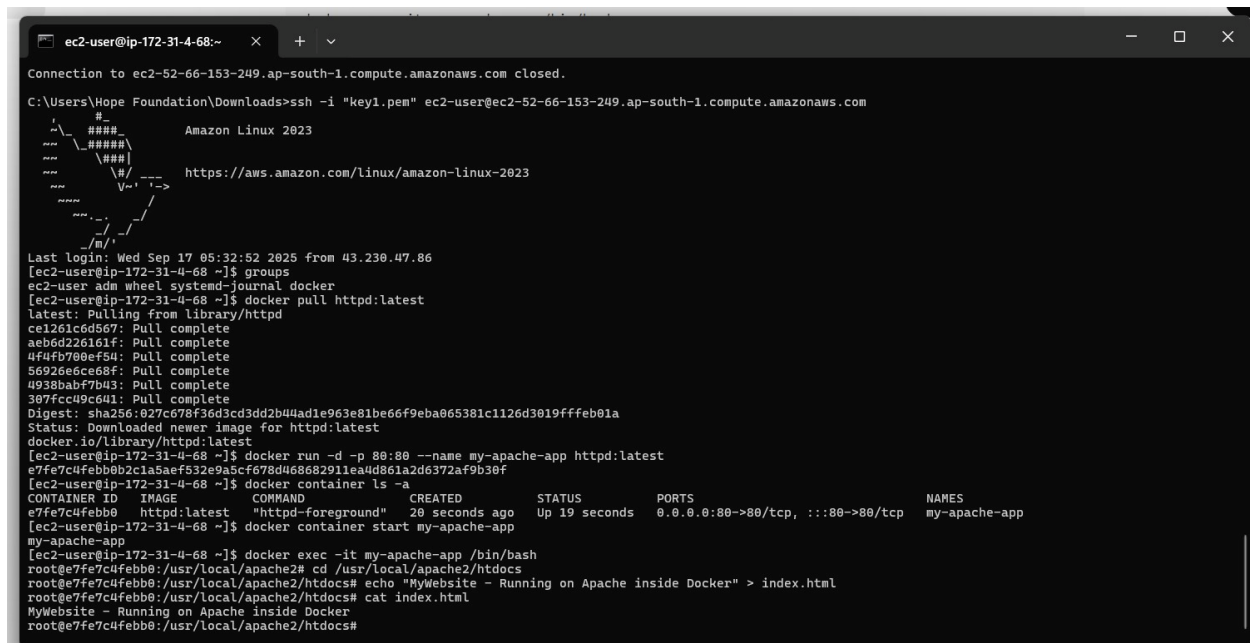




Step 2: Connect to EC2 via SSH

From your terminal:

```
ssh -i your-key.pem ec2-user@<EC2-Public-IP>
```



Step 3: Install Docker on Amazon Linux 2

Update and upgrade system:

```
sudo yum update -y
```

```
sudo yum upgrade -y
```

Install utilities:

```
sudo yum install -y yum-utils
```

Enable and install Docker:

```
sudo amazon-linux-extras enable docker
```

```
sudo yum install -y docker
```

Check Docker version:

```
docker --version
```

Start and enable Docker:

```
sudo service docker start
```

```
sudo systemctl enable docker
```

```
sudo systemctl status docker
```

(Optional) Add ec2-user to Docker group:

```
sudo usermod -aG docker ec2-user
```

```
exit
```

Reconnect via SSH:

```
ssh -i your-key.pem ec2-user@<EC2-Public-IP>
```

```
groups
```

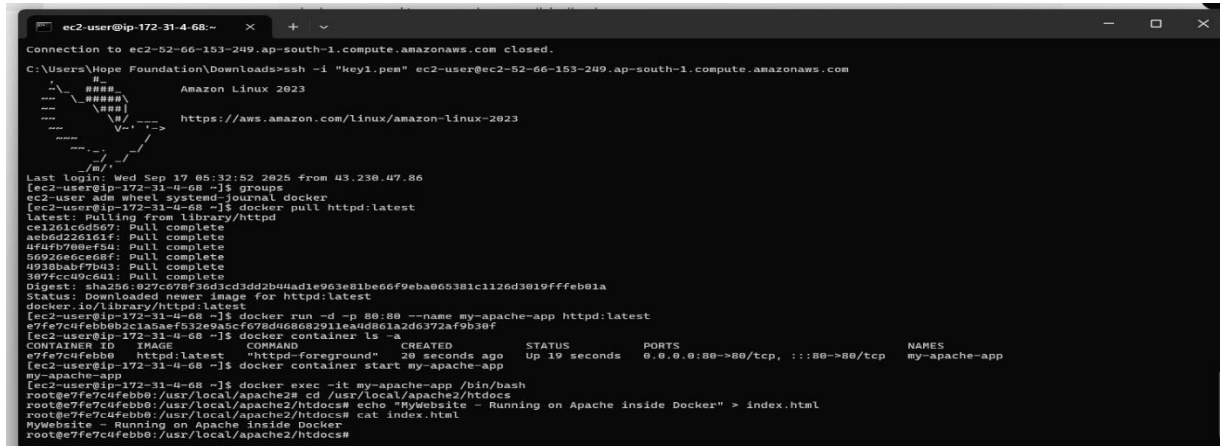


```
echo "MyWebsite - Running on Apache inside Docker" > index.html
```

cat index.html

Exit container:

exit



```
ec2-user@ip-172-31-4-68:~  
Connection to ec2-52-66-153-249.ap-south-1.compute.amazonaws.com closed.  
C:\Users\Hope Foundation\Downloads>ssh -i "key1.pem" ec2-user@ec2-52-66-153-249.ap-south-1.compute.amazonaws.com  
Amazon Linux 2023  
https://aws.amazon.com/linux/amazon-linux-2023  
Last login: Wed Sep 17 05:32:52 2025 from 43.230.47.86  
[ec2-user@ip-172-31-4-68 ~]$ groups  
ec2-user adm wheel systemd-journal docker  
[ec2-user@ip-172-31-4-68 ~]$ docker pull httpd:latest  
latest: Pulling from library/httpd  
ce1261cdd507: Pull complete  
a8b8d22d161f: Pull complete  
4f4b700ef54: Pull complete  
5692666ce68f: Pull complete  
4939babf7b43: Pull complete  
307fcc49c641: Pull complete  
Digest: sha256:827c078f36d3cd3dd2b04ad1e963e81be66f9eba065381c1126d3019fffeb01a  
Status: Downloaded newer image for httpd:latest  
docker.io/library/httpd:latest  
[ec2-user@ip-172-31-4-68 ~]$ docker run -d -p 80:80 --name my-apache-app httpd:latest  
e7fe7c4febb0b2c1a5aef632e9a5cf678d46860291lea4d861a2d6372af9b30f  
[ec2-user@ip-172-31-4-68 ~]$ docker container ls -a  
CONTAINER ID   IMAGE      COMMAND                  CREATED    STATUS    PORTS                               NAMES  
e7fe7c4febb0   httpd:latest  "httpd-foreground"      20 seconds ago    Up 19 seconds    0.0.0.0:80->80/tcp, :::80->80/tcp    my-apache-app  
[ec2-user@ip-172-31-4-68 ~]$ docker container start my-apache-app  
my-apache-app  
[ec2-user@ip-172-31-4-68 ~]$ docker exec -it my-apache-app /bin/bash  
root@e7fe7c4febb0:/usr/local/apache2# cd /usr/local/apache2/htdocs  
root@e7fe7c4febb0:/usr/local/apache2/htdocs# echo "MyWebsite - Running on Apache inside Docker" > index.html  
root@e7fe7c4febb0:/usr/local/apache2/htdocs# cat index.html  
MyWebsite - Running on Apache inside Docker  
root@e7fe7c4febb0:/usr/local/apache2/htdocs#
```

Step 6: Test the Web App

From your local machine:

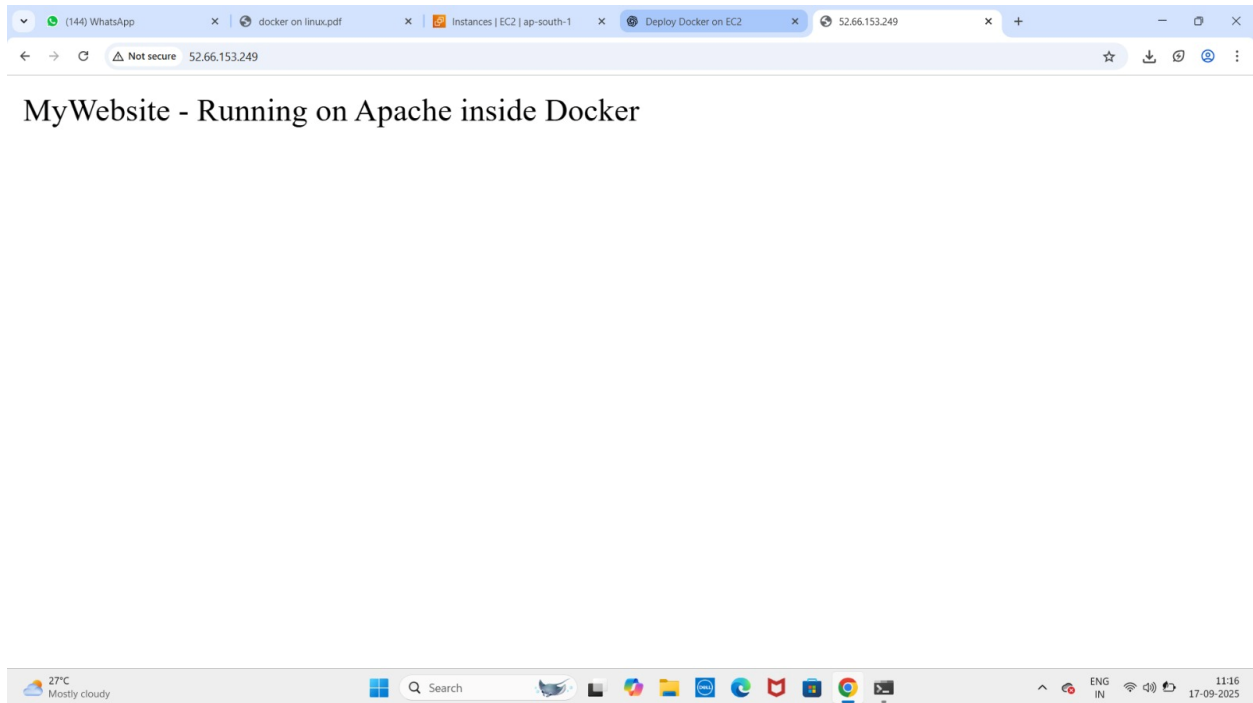
curl http://<EC2-Public-IP>

Or open in browser:

http://<EC2-Public-IP>

You should see:

MyWebsite - Running on Apache inside Docker



Step 7: Manage and Monitor

Inspect container:

```
docker container inspect my-apache-app
```

Check performance:

```
docker container stats my-apache-app
```

